

OPTIMIZING PARKING DISTRIBUTION IN HONG KONG

PRESENTED BY
SEARCH EVERYTHING



PROBLEM ANALYSIS

Major parking problems in Hong Kong:

1

**SEVERE
SHORTAGE &
UNEVEN
DISTRIBUTION**

2

**LACK OF A
UNIFIED
INFORMATION
PLATFORM**

3

**UPCOMING
SOUTHBOUND
TRAVEL SCHEME**

4

**CONTINUOUSLY
RISING ON-
STREET PARKING
FEES**





PROJECT OBJECTIVES

1

CREATE AN
INTEGRATED
PLATFORM for
PARKING METERS'
LOCATION &
INFORMATION

2

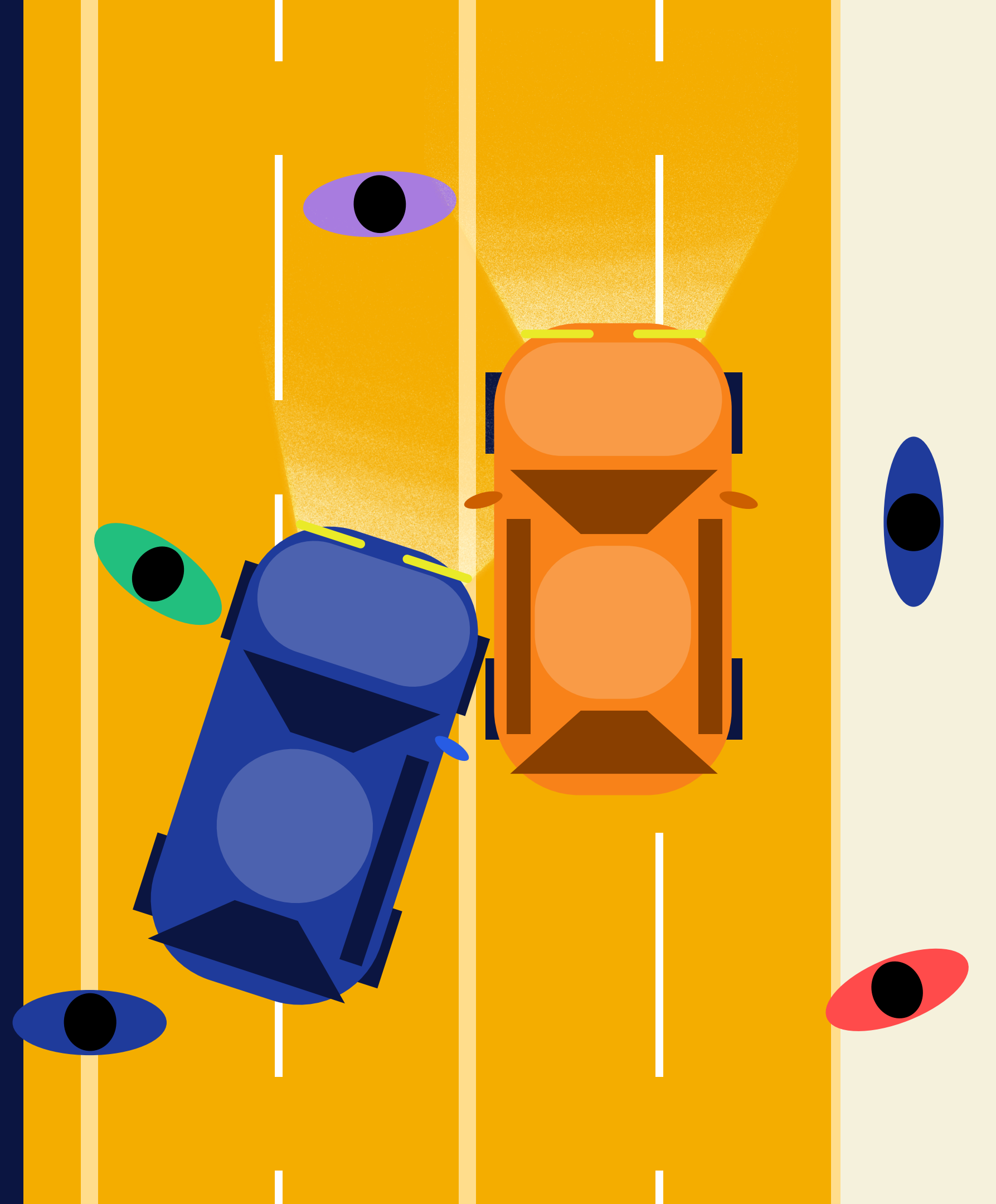
PROVIDES
LOCATION &
INFORMATION OF
CHARGING SPOTS.

3

USER REPORTS
COST OF
CHARGING
SPOTS.

4

VIEWING DENSITY
OF PARKING
METERS AND
CHARGING SPOTS.





DATASET SOURCE

1 DISTRIBUTION OF METERED PARKING SPACES
[TRANSPORT DEPARTMENT]

2 PUBLIC ELECTRIC VEHICLE CHARGERS LIST
[ENVIRONMENTAL PROTECTION DEPARTMENT]

3 HONG KONG 18 DISTRICTS BOUNDARY
[COMMON SPATIAL DATA INFRASTRUCTURE]



OUR TECHNICAL APPROACH

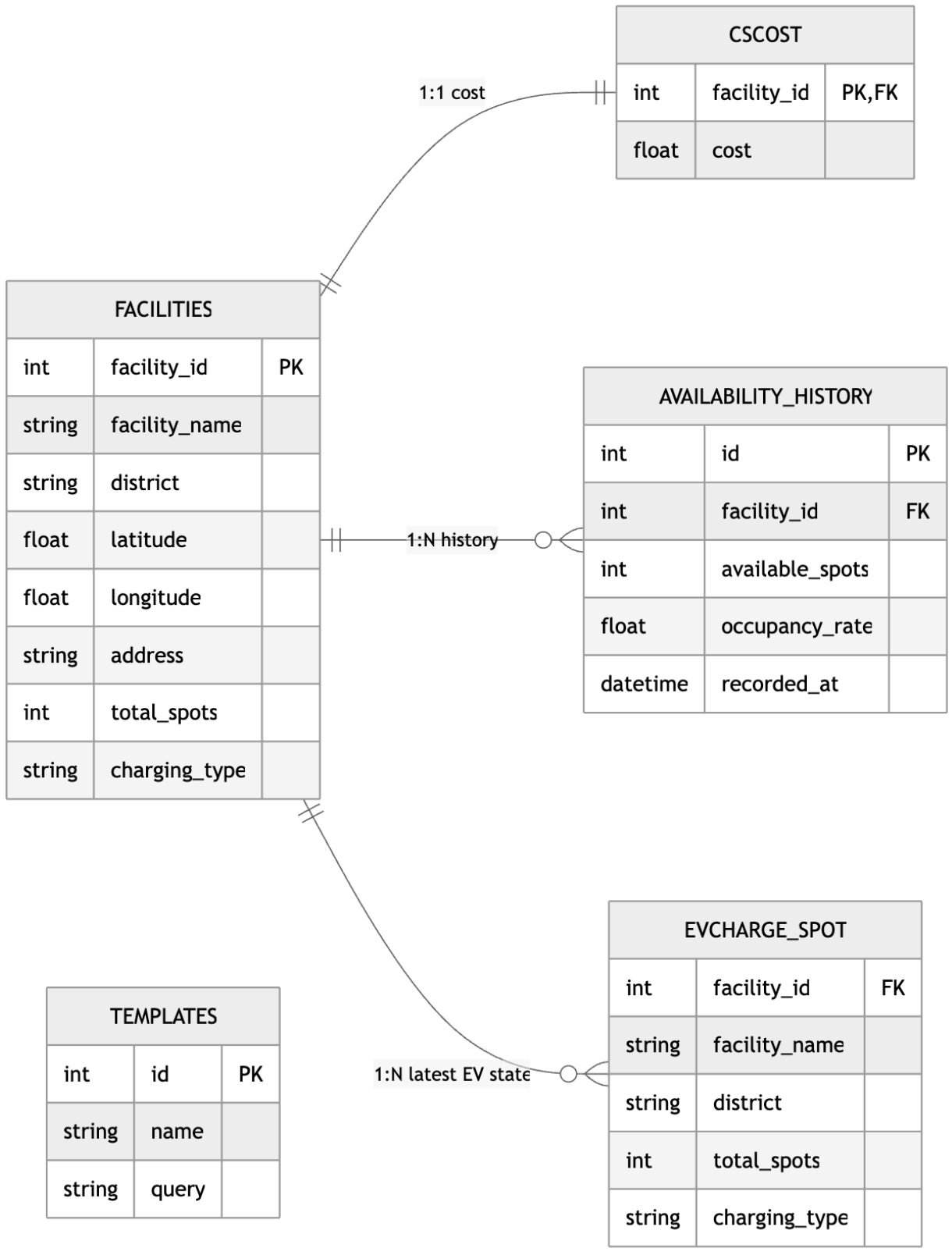


Table: templates				
	id	name	description	
...	Filter		Filter	Filter
1	1	Find Parking Meters Location	Find all on-street metered parking spaces in {district} for ...	SELECT ...
2	2	Find EV Charging Location By District	Currently {available_spots} out of {total_spots} charging ...	SELECT ...
3	3	Find Cost of EV chargespot	Charging fee in {district} district: {charging_type} charger...	SELECT ...
4	4	Report Cost of EV chargespot	Successfully updated charging cost to HK\${new_cost} for ...	UPDATE availability_histor
5	5	Find All Records in our database	Database contains {total_facilities} facilities ...	SELECT...

This database consists of five tables. The **facilities** table uses **facility_id** as its primary key and serves as the central entity.

The **cscost** table uses **facility_id** as both its primary key and foreign key that references **facilities.facility_id**, thereby establishing a one-to-one relationship with facilities. The **availability_history** table uses **id** as its primary key and employs **facility_id** as a foreign key that references **facilities.facility_id**, thereby establishing a one-to-many relationship (one facility to many historical records).

We designed a normalized database with 2 core tables to store all parking and EV charging data.

- **AVAILABILITY_HISTORY**
- **FACILITIES**
- **CSCOST**
- **EVCHARGE_SPOT**

ADDITIONALLY

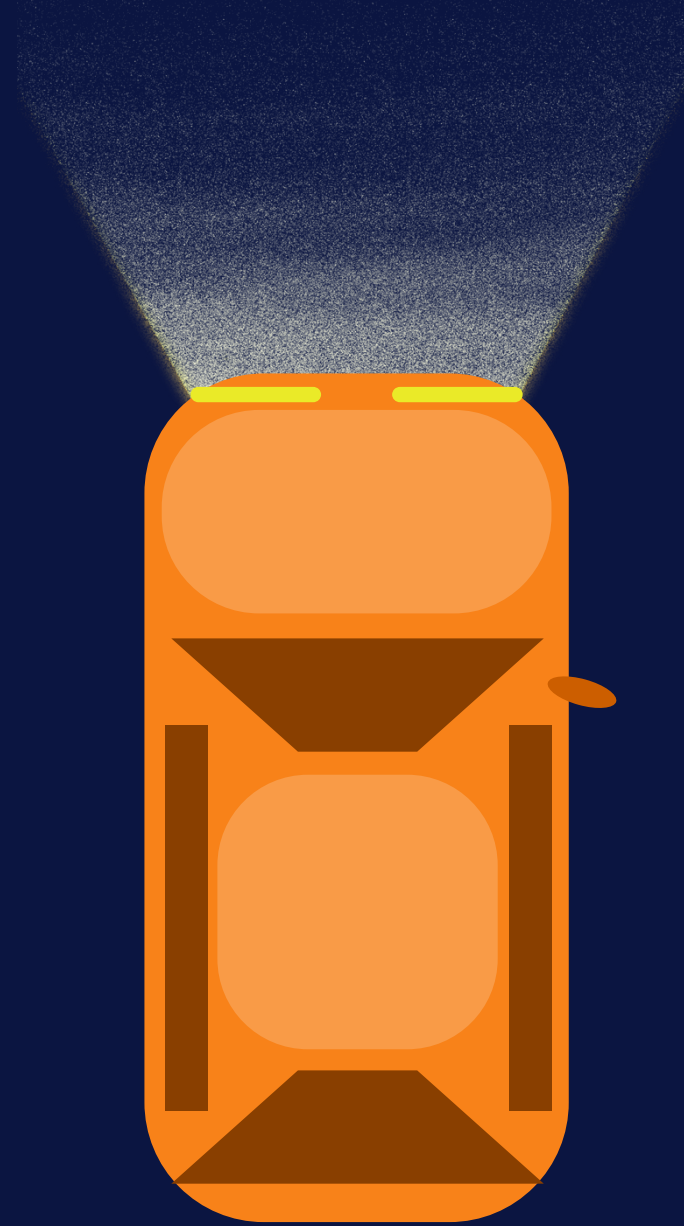
We included one more tables to store queries and natural-language report templates directly as data:

- **Templates**

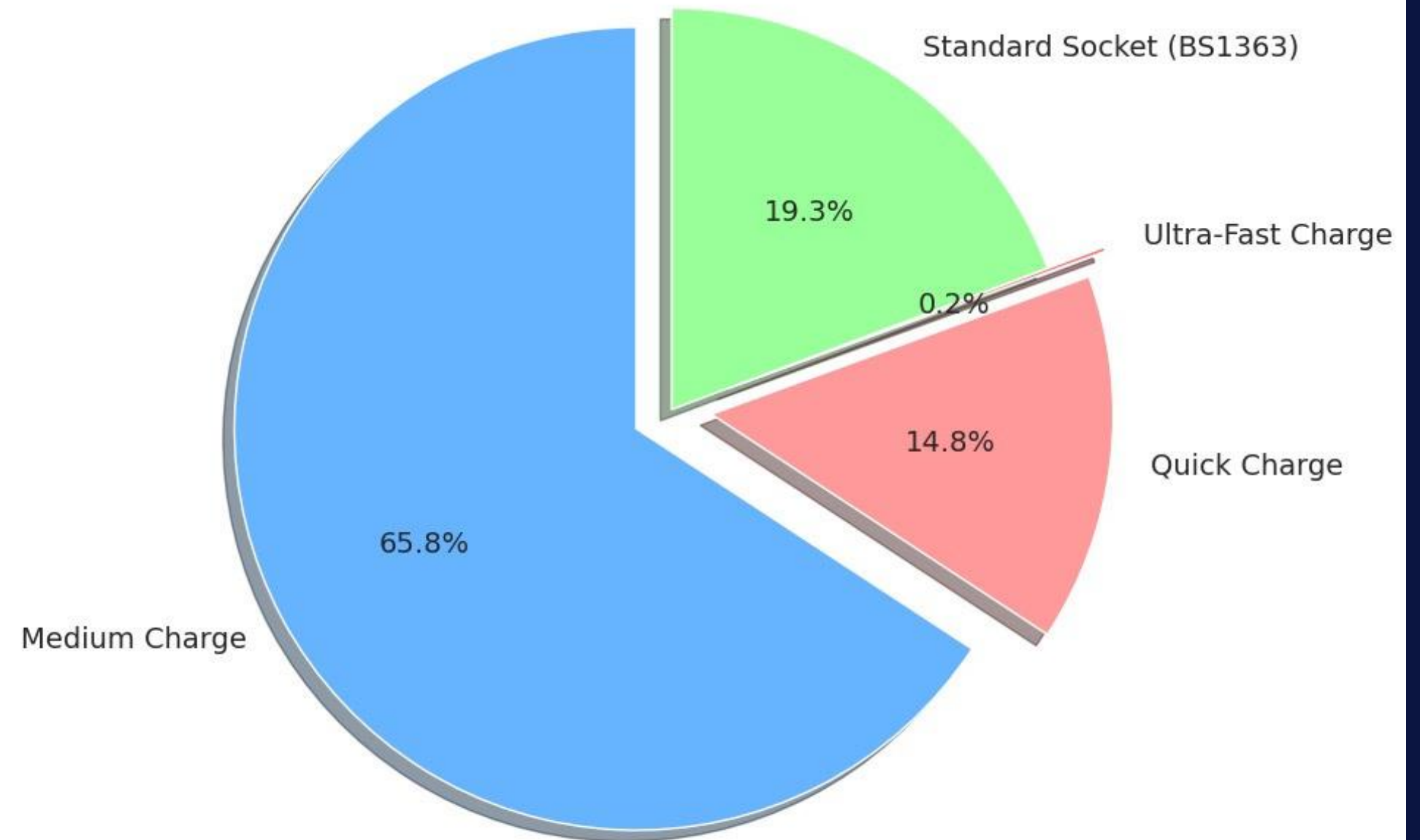
The **evcharge_spot** table employs **facility_id** as a foreign key that references **facilities.facility_id** (without declaring its own primary key), thereby functioning as a current-state snapshot or materialized view for EV charging facilities. The **templates** table uses **id** as its primary key and has no foreign keys, remaining completely independent.

BREAKDOWN OF PUBLIC EV CHARGERS BY CHARGING SPEED

**CHARGER
BREAKDOWN →
MEDIUM: 10,746 |
QUICK: 2,420 |
ULTRA-FAST: 27 |
STANDARD: 3,149**



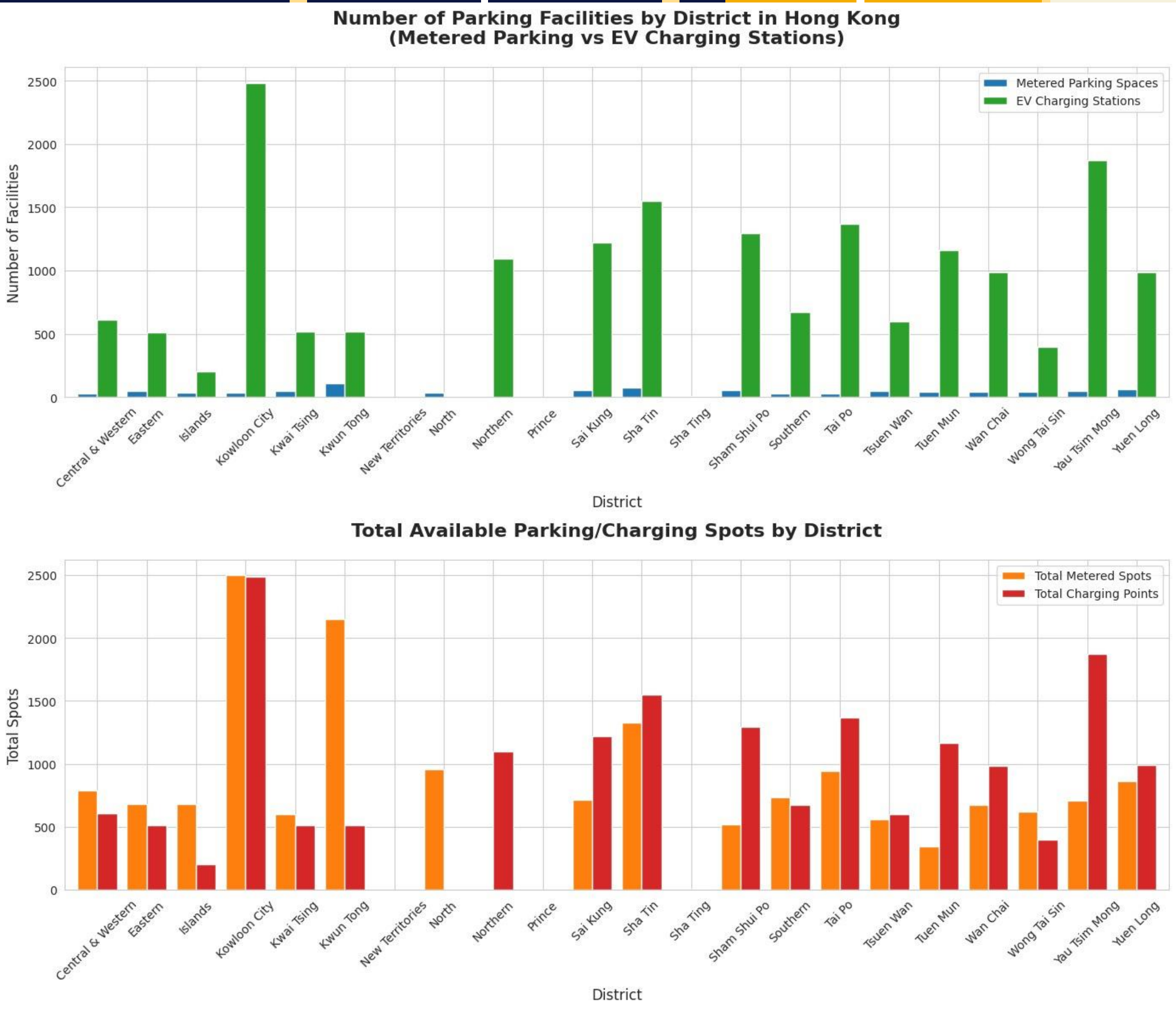
Distribution of EV Charger Types in Hong Kong (Latest 2025 Data)



FROM METERED PARKING TO EV READINESS – DISTRICT INSIGHTS

Number of Parking Facilities by District in Hong Kong (Metered Parking vs EV Charging Stations)

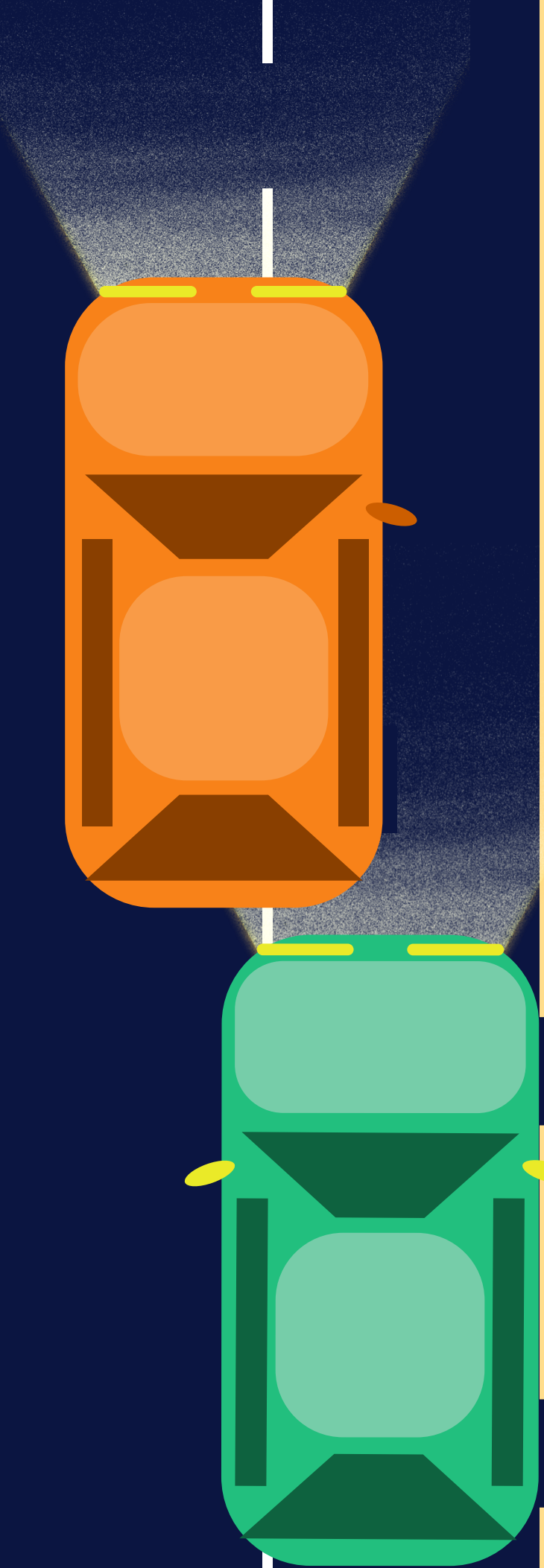
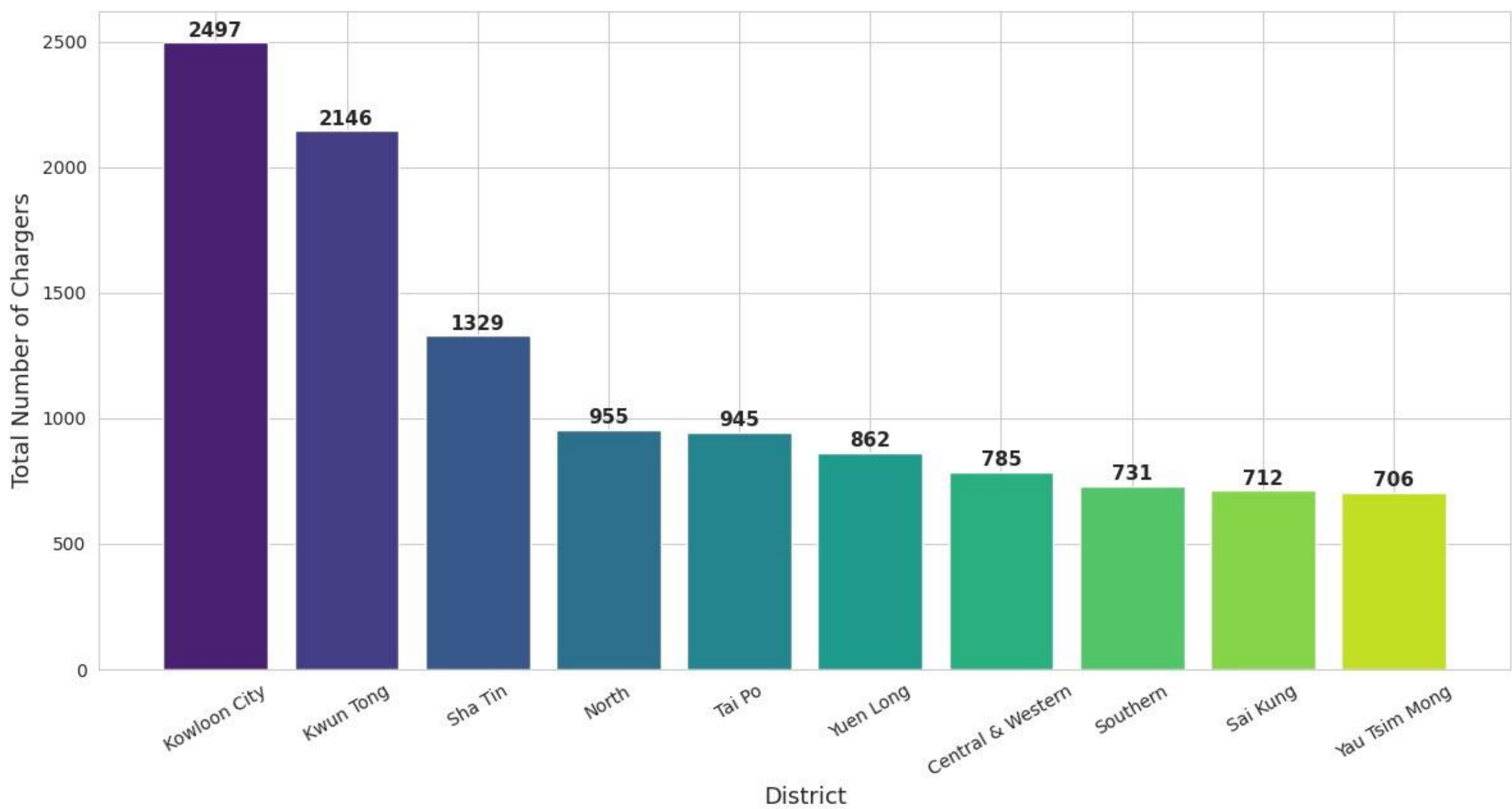
Total Available Parking/Charging Spots by District

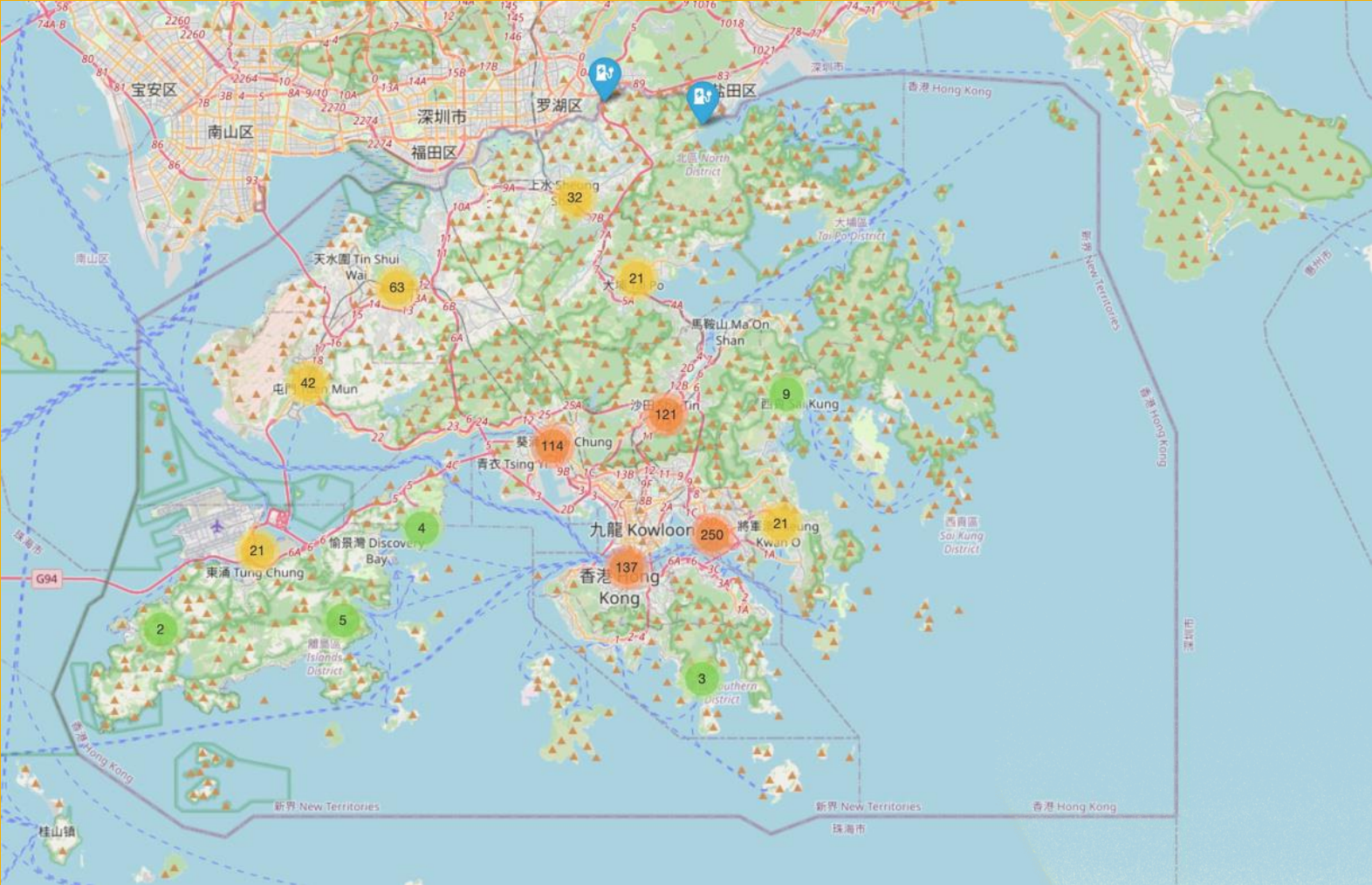
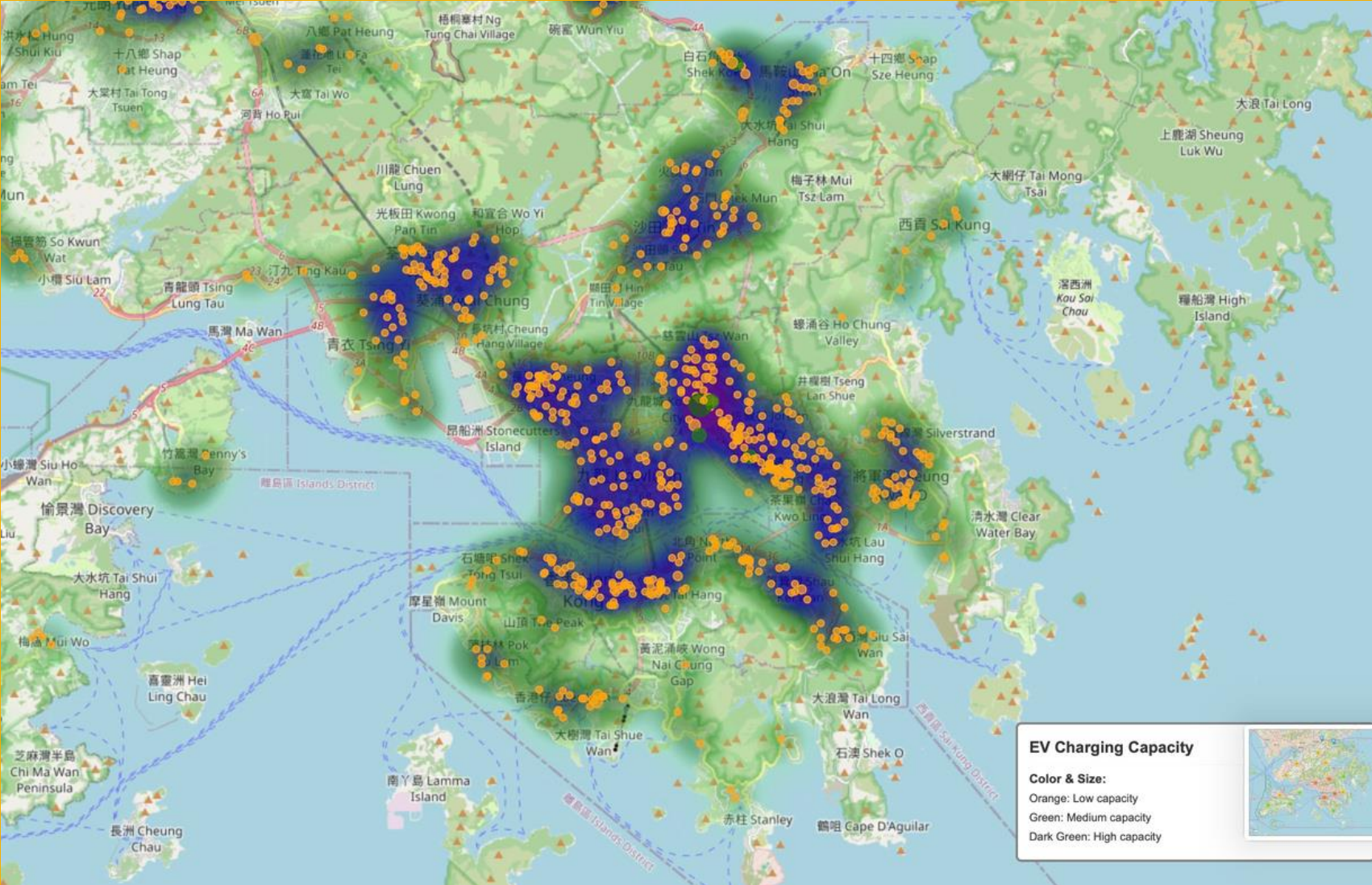


TOP 10 DISTRICTS WITH THE MOST EV CHARGING POINTS

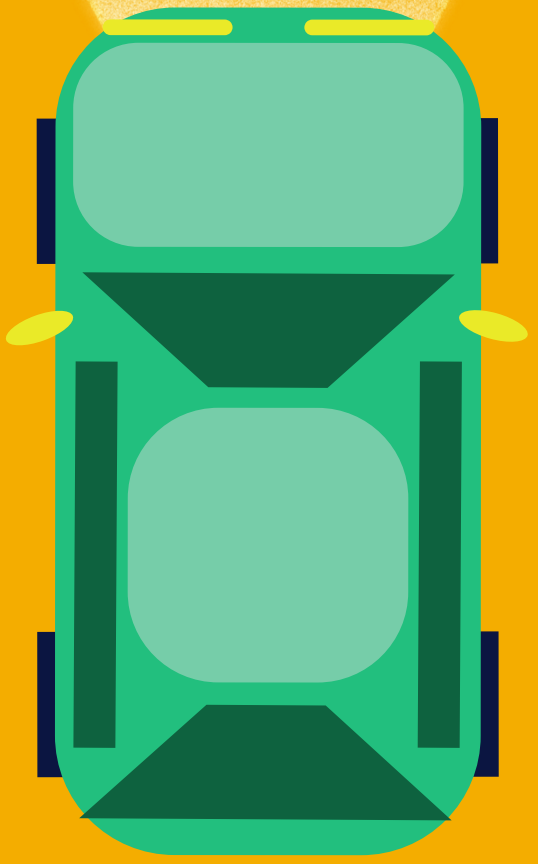
Kowloon City leads with 2,497 public charging points

Top 10 Districts with the Most EV Charging Points in Hong Kong





EV CHARGING NETWORK





LIMITATIONS OF CURRENT STUDY

LIMITATION	EXPLANATION
1. OCCUPANCY RATE SHOWS ONLY 0% OR 50%	1. THESE ARE PRESET PLACEHOLDER VALUES IN THE ORIGINAL DATASET USED FOR DEMONSTRATION PURPOSES ONLY.
2. NO GENUINE REAL-TIME AVAILABILITY DATA	2. WE COULD NOT FETCH LIVE OCCUPANCY DATA DURING THIS PROJECT.
3. ANALYSIS BASED ON A SINGLE SNAPSHOT	3. HISTORICAL TRENDS AND PEAK/OFF-PEAK PATTERNS ARE NOT YET AVAILABLE.
4. SOME DISTRICTS APPEAR TO HAVE ZERO CHARGERS	4. REFLECTS THE ACTUAL 2025 EPD DATASET; NO DATA SUPPLEMENTATION WAS PERFORMED.



KEY FINDINGS

COMPREHENSIVE PARKING & CHARGING LOCATOR

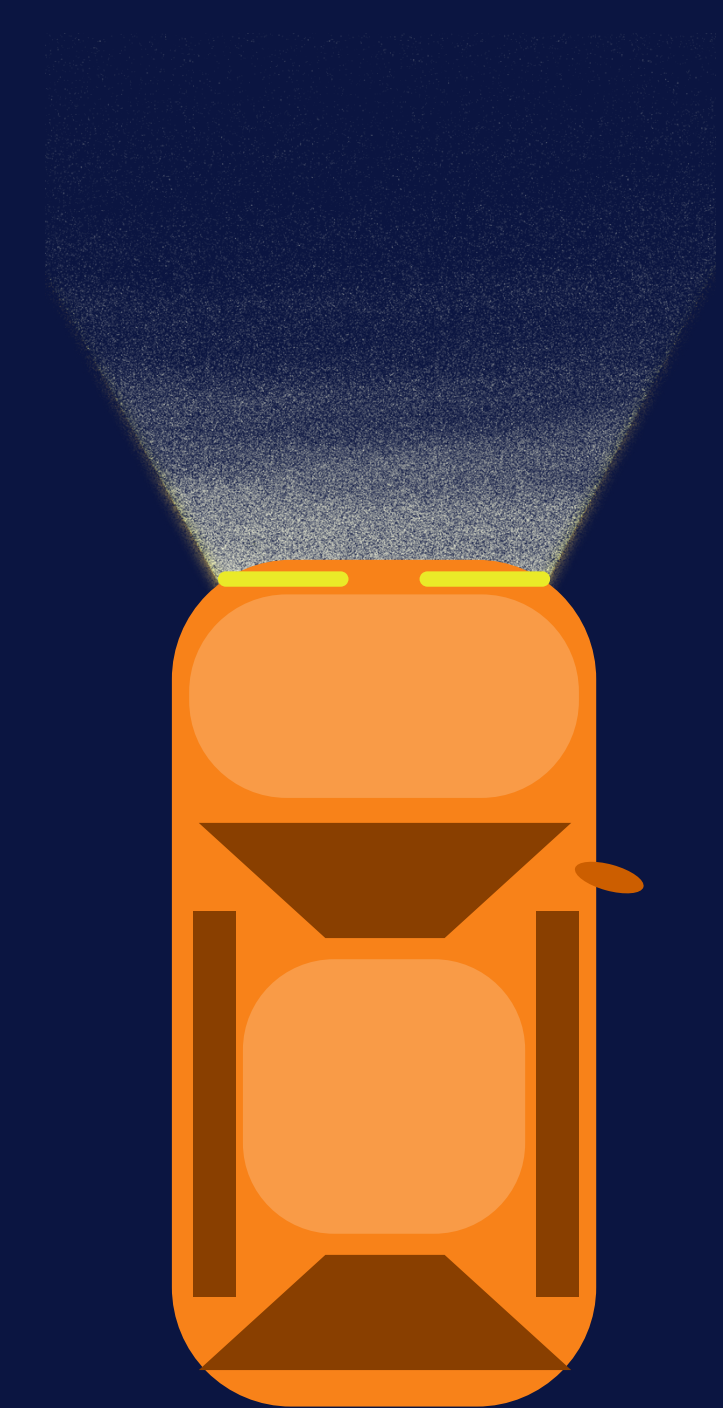
- The platform enables users to search for various types of metered parking locations across Hong Kong, including service hours and availability — supported by both data tables and map visualization.

EV Charging Meter Identification

- Users can locate metered parking spots equipped with electric vehicle (EV) charging capabilities. Future versions will integrate user-reported pricing data for enhanced transparency.

User-Driven Feedback System

- A built-in feedback mechanism allows users to submit real-time updates on charging costs, helping to keep the pricing information accurate and community-maintained.

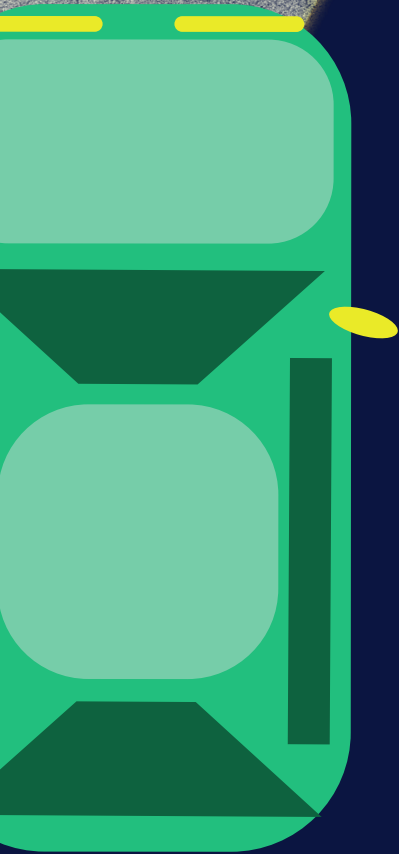




CONCLUSION

The current platform offers a functional and user-centric tool for locating both general and EV-specific metered parking in Hong Kong.

By combining clear data presentation with interactive maps and incorporating a user feedback system for pricing updates, it lays a strong foundation for supporting the city's transition toward electric mobility and smarter urban navigation.



FUTURE WORK

1

FOCUS NEW INFRASTRUCTURE ON UNDERSERVED DISTRICTS (ISLANDS, TSUEN WAN, YUEN LONG, ETC.)

2

INTEGRATE LIVE APIS TO POPULATE REAL-TIME AVAILABILITY IN THE AVAILABILITY_HISTORY TABLE

3

DEVELOP A UNIFIED MOBILE APP/DASHBOARD SHOWING BOTH PARKING AND CHARGING REAL-TIME STATUS

4

BUILD TIME-SERIES FORECASTING MODELS FOR OCCUPANCY PREDICTION



THANK YOU !

